

Q1 - 24 January - Shift 1

Let $p, q \in \mathbb{R}$ and $(1 - \sqrt{3}i)^{200} = 2^{199}(p + iq)$,

Space for your notes:

$i = \sqrt{-1}$ Then $p + q + q^2$ and $p - q + q^2$ are roots of the equation.

(1) $x^2 + 4x - 1 = 0$ (2) $x^2 - 4x + 1 = 0$

(3) $x^2 + 4x + 1 = 0$ (4) $x^2 - 4x - 1 = 0$

Q2 - 24 January - Shift 2

The value of $\left(\frac{1 + \sin \frac{2\pi}{9} + i \cos \frac{2\pi}{9}}{1 + \sin \frac{2\pi}{9} - i \cos \frac{2\pi}{9}} \right)^3$ is

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(1) $-\frac{1}{2}(1 - i\sqrt{3})$

(2) $\frac{1}{2}(1 - i\sqrt{3})$

(3) $-\frac{1}{2}(\sqrt{3} - i)$

(4) $\frac{1}{2}(\sqrt{3} + i)$

Q3 - 25 January - Shift 1

Let $z_1 = 2 + 3i$ and $z_2 = 3 + 4i$. The set

$$S = \left\{ z \in \mathbb{C} : |z - z_1|^2 - |z - z_2|^2 = |z_1 - z_2|^2 \right\}$$

represents a

- (1) straight line with sum of its intercepts on the coordinate axes equals 14
- (2) hyperbola with the length of the transverse axis 7
- (3) straight line with the sum of its intercepts on the coordinate axes equals -18
- (4) hyperbola with eccentricity 2

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Q4 - 25 January - Shift 2

Let z be a complex number such that

$$\left| \frac{z - 2i}{z + i} \right| = 2, z \neq -i. \text{ Then } z \text{ lies on the circle of}$$

radius 2 and centre

- (1) (2, 0)
- (2) (0, 0)
- (3) (0, 2)
- (4) (0, -2)

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Q5 - 29 January - Shift 1

For two non-zero complex number z_1 and z_2 , if $\operatorname{Re}(z_1 z_2) = 0$ and $\operatorname{Re}(z_1 + z_2) = 0$, then which of the following are possible ?

(A) $\operatorname{Im}(z_1) > 0$ and $\operatorname{Im}(z_2) > 0$

(B) $\operatorname{Im}(z_1) < 0$ and $\operatorname{Im}(z_2) > 0$

(C) $\operatorname{Im}(z_1) > 0$ and $\operatorname{Im}(z_2) < 0$

(D) $\operatorname{Im}(z_1) < 0$ and $\operatorname{Im}(z_2) < 0$

Choose the correct answer from the options given

below :

(1) B and D

(2) B and C

(3) A and B

(4) A and C

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Q6 - 29 January - Shift 2

Let $\alpha = 8 - 14i$, $A = \left\{ z \in \mathbb{C} : \frac{\alpha z - \bar{\alpha} \bar{z}}{z^2 - (\bar{z})^2 - 112i} = 1 \right\}$

and $B = \{ z \in \mathbb{C} : |z + 3i| = 4 \}$.

Then $\sum_{z \in A \cap B} (\operatorname{Re} z - \operatorname{Im} z)$ is equal to _____.

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Q7 - 30 January - Shift 1

Let $z = 1 + i$ and $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$. Then $\frac{12}{\pi}$

$\arg(z_1)$ is equal to _____.

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Q8 - 31 January - Shift 1

For all $z \in \mathbb{C}$ on the curve $C_1 : |z| = 4$, let the locus

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of the point $z + \frac{1}{z}$ be the curve C_2 . Then

- (1) the curves C_1 and C_2 intersect at 4 points
- (2) the curves C_1 lies inside C_2
- (3) the curves C_1 and C_2 intersect at 2 points
- (4) the curves C_1 lies inside C_2

Q9 - 31 January - Shift 2

The complex number $z = \frac{i-1}{\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}}$ is equal

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to:

- (1) $\sqrt{2} \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)$
- (2) $\cos \frac{\pi}{12} - i \sin \frac{\pi}{12}$
- (3) $\sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$
- (4) $\sqrt{2} i \left(\cos \frac{5\pi}{12} - i \sin \frac{5\pi}{12} \right)$

Q10 - 01 February - Shift 1

If the center and radius of the circle $\left| \frac{z-2}{z-3} \right| = 2$ are respectively (α, β) and γ , then $3(\alpha + \beta + \gamma)$ is equal to

- (1) 11
- (2) 9
- (3) 10
- (4) 12

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Q11 - 01 February - Shift 2

Let a, b be two real numbers such that $ab < 0$. If the complex number $\frac{1+ai}{b+i}$ is of unit modulus and $a+ib$ lies on the circle $|z-1|=|2z|$, then a possible value of $\frac{1+[a]}{4b}$, where $[t]$ is greatest integer

function, is :

- (1) $-\frac{1}{2}$
- (2) -1
- (3) 1
- (4) $\frac{1}{2}$

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Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2)

Q2 (3)

Q3 (1)

Q4 (4)

Q5 (2)

Q6 (14)

Q7 (9)

Q8 (1)

Q9 (1)

Q10 (4)

Q11 (1)

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Q1 (2)

$$(1 - \sqrt{3}i)^{200} = 2^{199}(p + iq)$$

$$2^{200} \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^{200} = 2^{199}(p + iq)$$

$$2 \left(-\frac{1}{2} - i \frac{\sqrt{3}}{2} \right) = p + iq$$

$$p = -1, q = -\sqrt{3}$$

$$\alpha = p + q + q^2 = 2 - \sqrt{3}$$

$$\beta = p - q + q^2 = 2 + \sqrt{3}$$

$$\alpha + \beta = 4$$

$$\alpha \cdot \beta = 1$$

$$\text{equation } x^2 - 4x + 1 = 0$$

Q2 (3)

$$\text{Let } \sin \frac{2\pi}{9} + i \cos \frac{2\pi}{9} = z$$

$$\left(\frac{1+z}{1+\bar{z}} \right)^3 = \left(\frac{1+z}{1+\frac{1}{z}} \right)^3 = z^3$$

$$\Rightarrow \left(i \left(\cos \frac{2\pi}{9} - i \sin \frac{2\pi}{9} \right) \right)^3$$

$$= -i \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right) = -i \left(\frac{-1}{2} - i \frac{\sqrt{3}}{2} \right)$$

$$\Rightarrow \frac{-1}{2} (\sqrt{3} - i).$$

Q3 (1)

$$\left((x-2)^2 + (y-3)^2 \right) - \left((x-3)^2 + (y-4)^2 \right) = 1+1$$

$$\Rightarrow x+y=7$$

Q4 (4)

$$(z-2i)(\bar{z}+2i) = 4(z+i)(\bar{z}-i)$$

$$z\bar{z} + 4 + 2i(z-\bar{z}) = 4(z\bar{z} + 1 + i(\bar{z}-z))$$

$$3z\bar{z} - 6i(z-\bar{z}) = 0$$

$$x^2 + y^2 - 2i(2iy) = 0$$

$$x^2 + y^2 + 4y = 0$$

Q5 (2)

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$$z_1 = x_1 + iy_1$$

$$z_2 = x_2 + iy_2$$

$$\operatorname{Re}(z_1 z_2) = x_1 x_2 - y_1 y_2 = 0$$

$$\operatorname{Re}(z_1 + z_2) = x_1 + x_2 = 0$$

x_1 & x_2 are of opposite sign

y_1 & y_2 are of opposite sign

Q6 (14)

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$$\alpha = 8 - 14i$$

$$z = x + iy$$

$$az = (8x + 14y) + i(-14x + 8y)$$

$$z + \bar{z} = 2x \quad z - \bar{z} = 2iy$$

$$\text{Set A: } \frac{2i(-14x + 8y)}{i(4xy - 112)} = 1$$

$$(x - 4)(y + 7) = 0$$

$$x = 4 \quad \text{or} \quad y = -7$$

$$\text{Set B: } x^2 + (y + 3)^2 = 16$$

$$\text{when } x = 4 \quad y = -3$$

$$\text{when } y = -7 \quad x = 0$$

$$\therefore A \cap B = \{4 - 3i, 0 - 7i\}$$

$$\text{So, } \sum_{z \in A \cap B} (\operatorname{Re} z - \operatorname{Im} z) = 4 - (-3) + (0 - (-7)) = 14$$

Q7 (9)

$$z = 1 + i$$

$$z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$$

$$z_1 = \frac{1 + i(1 - i)}{(1 - i)(1 - 1 - i) + \frac{1}{1 + i}}$$

$$= \frac{1 + i - i^2}{(1 - i)(-i) + \frac{1 - i}{2}}$$

$$= \frac{2 + i}{-3i - 1} = \frac{4 + 2i}{-3i - 1}$$

$$= \frac{-(4 + 2i)(3i - 1)}{(3i)^2 - (1)^2}$$

$$\text{Arg}(z_1) = \frac{3\pi}{4}$$

$$\therefore \frac{12}{\pi} \arg(z_1) = \frac{12}{\pi} \times \frac{3\pi}{4} = 9$$

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Q8 (1)

$$\text{Let } w = z + \frac{1}{z} = 4e^{i\theta} + \frac{1}{4}e^{-i\theta}$$

$$\Rightarrow w = \frac{17}{4}\cos\theta + i\frac{15}{4}\sin\theta$$

$$\text{So locus of } w \text{ is ellipse } \frac{x^2}{\left(\frac{17}{4}\right)^2} + \frac{y^2}{\left(\frac{15}{4}\right)^2} = 1$$

$$\text{Locus of } z \text{ is circle } x^2 + y^2 = 16$$

Q9 (1)

$$Z = \frac{i-1}{\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}} = \frac{i-1}{\frac{1}{2} + \frac{\sqrt{3}}{2}i}$$

$$= \frac{i-1}{\frac{1}{2} + \frac{\sqrt{3}}{2}i} \times \frac{\frac{1}{2} - \frac{\sqrt{3}}{2}i}{\frac{1}{2} - \frac{\sqrt{3}}{2}i} = \frac{\sqrt{3}-1}{2} + \frac{\sqrt{3}+1}{2}i$$

Apply polar form,

$$r \cos \theta = \frac{\sqrt{3}-1}{2}$$

$$r \sin \theta = \frac{\sqrt{3}+1}{2}$$

Now, $\tan \theta = \frac{\sqrt{3}+1}{\sqrt{3}-1}$

$$\theta = \frac{5\pi}{12}$$

Q10 (4)

$$\begin{aligned}
 \sqrt{(x-2)^2 + y^2} &= 2\sqrt{(x-3)^2 + y^2} \\
 x^2 + y^2 - 4x + 4 &= 4x^2 + 4y^2 - 24x + 36 \\
 3x^2 + 3y^2 - 20x + 32 &= 0 \\
 x^2 + y^2 - \frac{20}{3}x + \frac{32}{3} &= 0 \\
 (\alpha, \beta) &= \left(\frac{10}{3}, 0\right) \\
 \gamma &= \sqrt{\frac{100}{9} - \frac{32}{3}} = \sqrt{\frac{4}{9}} = \frac{2}{3} \\
 3(\alpha, \beta, \gamma) &= 3\left(\frac{10}{3} + \frac{2}{3}\right) \\
 &= 12
 \end{aligned}$$

Q11 (1)

$$ab < 0 \quad \left| \frac{1+ai}{b+i} \right| = 1$$

$$|1+ai| = |b+i|$$

$$a^2 + 1 = b^2 + 1 \Rightarrow a = \pm b \Rightarrow b = -a \quad \text{as } ab < 0$$

$$(a, b) \text{ lies on } |z-1| = |2z|$$

$$|a+ib-1| = 2|a+ib|$$

$$(a-1)^2 + b^2 = 4(a^2 + b^2)$$

$$(a-1)^2 = a^2 = 4(2a^2)$$

$$1-2a = 6a^2 \Rightarrow 6a^2 + 2a - 1 = 0$$

$$a = \frac{-2 \pm \sqrt{28}}{12} = \frac{-1 \pm \sqrt{7}}{6}$$

$$a = \frac{\sqrt{7}-1}{6} \text{ \& } b = \frac{1-\sqrt{7}}{6}$$

$$[a] = 0$$

$$\therefore \frac{1+[a]}{4b} = \frac{6}{4(1-\sqrt{7})} = -\left(\frac{1+\sqrt{7}}{4}\right)$$

$$\text{or } [a] = 0$$

$$\text{Similarly it is not matching with } a = \frac{-1-\sqrt{7}}{6}$$

No answer is matching.